Mushfique Shahriar Aurnab

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Education

Daegu Gyeongbuk Institute of Science and Technology (DGIST) — Daegu, Republic of Korea
B.S. student intending to major in Computer Science and Electrical Engineering — *Aug. 2024 – Present*

Maastricht University, Center for European Studies — Maastricht, Netherlands
Summer Programme in European Studies — *Summer 2026*

- Courses: Global Governance of AI and Intercultural Communication.

Experience

Undergraduate Researcher, UGRP I — DGIST
Comparative Study of Catastrophic Forgetting in CNNs and Vision Transformers — *Spring 2026*

- Ran nine sequential CIFAR-10 experiments across three class splits and three random seeds. Trained linear probes at six depths to see where information from the first task remained after the model learned the second.
- After the model learned Task B, linear probes recovered 79.7% of the old-task examples that its final classifier got wrong. Retraining that classifier restored 74.6% mean old-task accuracy.
- Compared five ways to choose replay examples. Random selection performed best; none of the four targeted methods improved buffer-only recovery.

Computer Vision Lab, DGIST — Daegu, Republic of Korea
Student Researcher, Video Annotation — *May 2025 – Aug. 2025*

- Labeled video data for video-text alignment and text-based object detection experiments in Prof. Sunghoon Im's lab.

Personal Projects

GPX Motion Analysis | *Python, time-series alignment, Matplotlib* — 2026

- Built a command-line tool that aligns GPX tracks recorded at different times without filling long gaps or connecting separate track segments.
- Added CSV exports, motion metrics, stop-threshold comparisons, JSON summaries, and plots. The tests use reproducible synthetic tracks and run in CI on Python 3.11 and 3.14.

Vox Pop | *Python, MCP, FastEmbed* — 2026

- Built a Python library, command-line tool, and MCP server that searches nine public discussion sites. For each query, a semantic router selects the most relevant communities first.

Course Extractor | *Python, Requests, BeautifulSoup* — 2026

- Built a scraper that collects course listings from FOSE, JSON-LD, or HTML and saves them as JSON and Markdown. A Harvard DCE run found 508 courses without page errors.

MapleStory Code Battle AI | *Python, OpenCV, minimax* — 2025

- Reconstructed the game board from screen captures using template matching and color detection. Scored possible moves with summed-area tables and searched three plies ahead using minimax with alpha-beta pruning.

Technical Skills

Languages: Python, C, C++
Machine learning and vision: PyTorch, torchvision, scikit-learn, pandas, NumPy, OpenCV, Matplotlib, linear probing
Developer tools: Git, pytest, HTTP APIs, MCP servers, Python packaging, command-line tools, JSON, Markdown